

CS3014 - Estructura de Datos Avanzados

Final Project 2026-1

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1 Description and Tasks

The final project topic is free, as was mentioned in the last couple of lectures. You should be able to present a problem that requires implementing and applying at least one data structure. The data structures to be considered are not limited to those presented in the lectures. The project should include a visual component for a video demo.

The problem should not be trivial, like a Codeforces problem; it should be an application in a real-world problem (like 3D content similarity search engine, is like a Google before IA, but for each results it can display a 3D interface showing a graph with a screenshot of the nearest web pages similar in content, this could be a new way to navigate in a VR environment). Another possible direction could be a comparison study of different data structures or implementations, showing the results for a complete set of experiments (like, for instance, a complete extension of the *soft heap* project by showing all heaps' performance; this is not valid for the final project, choose another topic).

2 Deliverables and Submission

The deliverables for this project are the following:

1. The **GitHub project** for your code. Don't forget to add a **README** file that presents your project, lists the required dependencies, explains how to build and run it, includes a short description, and lists the authors.
2. Prepare a short presentation showing your project. The **presentation will be 15 minutes**; use the time wisely and **highlight your project's best features**.
3. Prepare a **demo video** showing the execution of your project.
4. You should submit a **short PDF report (2 to 3 pages, LaTeX)**, where you describe your work, experiments, results, and the contributions of each member, and the **link to the GitHub repository, demo video, and presentation**.
5. First deadline, **progress submission** (required for the final presentation) next **Friday, June 26, at 23:59 hrs**. It should be a draft of your project showing the current state and pending tasks.
6. The **project submission** will be available on Canvas, with a deadline of **Friday, July 10, at 00:00 hrs**. You should submit the PDF report with the required links.
7. The presentation will be on **Friday, July 10, at 13:00 hrs in room M1001**.
8. You can find the **grading rubric** on **Canvas**.